

Sina Moradi

RF / Microwave Engineer — PhD Researcher in Multiport RF Systems, Antenna Arrays, Noise, Photonics, and High-Frequency Modeling

Munich, Germany — Authorized to work in Germany — Open to relocation / international opportunities
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TUM Profile



Professional Summary

PhD researcher in High-Frequency Engineering at the Technical University of Munich with **thesis submission in progress**. My work spans **multiport RF systems, antenna arrays, beamforming, phased arrays, mutual coupling/matching, front-end mismatch, noise matching, correlated noise, and lossy transmission line effects**. I also have a strong background in **computational photonics**, including **anisotropic periodic multilayer structures, diffraction, LCoS / SLM modeling, and silicon photonics**. Additional experience includes **Cadence-based LNA/PA design, PCB design, RF measurements, OTA/chamber-based setups, and system-level simulation**. I am particularly interested in applied R&D roles in **RF systems and validation, phased-array / radar systems, high-frequency circuits, photonics / optical sensing, and advanced communication systems**.

Core Expertise

RF Systems & Arrays: multiport communication systems, antenna arrays, phased arrays, beamforming, mutual coupling, total active reflection coefficient (TARC), decoupling and matching networks, source/excitation optimization, end-to-end MIMO system thinking

Noise & Front-End Analysis: noise matching, correlated noise, front-end mismatching, lossy transmission-line noise, SNR / NF related analysis, receiver and transmitter chain behavior, high-impedance receiver concepts

Measurements & Validation: RF characterization, VNA / spectrum analyzer / signal generator / oscilloscope use, OTA and chamber-based setups, probe-station and near-field measurements / mmWave-style measurement exposure, calibration-sensitive workflows, structured post-processing and reporting

Circuits & Hardware: Cadence-based LNA/PA design, transistor-level RF simulation, PCB design, Altium Designer, KiCad, EMC-aware hardware implementation

Photonics & Wave Physics: anisotropic periodic multilayer structures, diffraction, Bloch–Floquet concepts, LCoS / SLM structures, silicon photonics, wave interaction in complex media

Additional Topics: reconfigurable intelligent surfaces (RIS), BAW filters, acousto-electromagnetic systems, radio-acoustic sensing concepts

Software & Simulation: MATLAB, Python, C++, Simulink, Keysight SystemVue, HFSS, CST, ADS, COMSOL, Altium Designer, KiCAD, FEKO, Momentum, Optenni Lab, Cadence, Lumerical, TCAD, LaTeX

Research & Professional Experience

Doctoral Research Assistant

Technical University of Munich (TUM), Chair of High-Frequency Engineering

Oct 2020 – Present

Munich, Germany

- Research on **multiport RF front-ends and antenna arrays**, including beamforming, matching, coupling reduction, scan-dependent performance, and source/excitation optimization.
- Investigated **noise matching, correlated noise, front-end mismatching, and lossy transmission-line effects** in high-frequency RF chains.
- Built quantitative **MATLAB/Python/SystemVue workflows** for RF system analysis, technical evaluation, and interpretation of measured/simulated data.
- Performed and supported **RF laboratory measurements**, calibration-sensitive evaluations, and reproducible documentation/post-processing workflows.

RF Engineer & J1 Visiting Faculty Researcher

Virginia Tech, Bradley Department of Electrical & Computer Engineering

Oct 2024 – May 2025

Blacksburg, VA, USA

- Conducted applied work on **front-end mismatching, mutual coupling, transmission-line noise, bandwidth, and Tx/Rx system trade-offs**.
- Worked with **Keysight SystemVue** on **end-to-end MIMO RF/baseband system modeling**, from transmit chain through physical channel to receiver.
- Identified and reported a **SystemVue modeling issue** to Keysight and discussed it directly with their engineers/developers.

Graduate Researcher

University of Calgary

Dec 2018 – Sep 2020

Calgary, Canada

- Designed, fabricated, and measured circularly polarized microstrip patch antennas.
- Built strong practical experience in **RF prototyping, antenna measurement, and network analysis**.

Selected Research Themes

- **Beamforming / phased arrays / multiport systems:** excitation-aware and excitation-independent characterization of coupled arrays; TARC, matching, scan behavior, and channel/system-level insight.
- **Physical elimination of decoupling and matching networks:** source-side / pre-distortion style realization of matching behavior in multiport antenna arrays.
- **Photonics / optical modeling:** M.Sc. thesis on **diffraction in multi-layer anisotropic periodic structures**, with relevance to LCoS / SLMs and wave-based sensing systems.
- **Advanced device/system topics:** RIS, BAW filters, acousto-electromagnetic systems, radio-acoustic sensing, silicon photonics, and RF–photonics-adjacent concepts.

Education

PhD Researcher, High-Frequency Engineering
Technical University of Munich

2020 – Present
 Munich, Germany

- Research focused on RF systems, antenna arrays, multiport networks, noise, beamforming, and high-frequency modeling.

M.Sc. Electrical Engineering (Communications – Field & Waves)
University of Tehran

2016 – 2018
 Tehran, Iran

- GPA: 18.27/20 — **Ranked 1st in the whole Faculty of Electrical & Computer Engineering.**
- Thesis: *Analysis of Diffraction by Multi-Layer Anisotropic Periodic Structures.*

B.Sc. Electrical Engineering, Communication Systems
Shahid Beheshti University

2012 – 2016
 Tehran, Iran

- GPA: 19.34/20 — **Ranked 1st in the whole Faculty of Electrical & Computer Engineering.**

Selected Publications & Manuscripts

- S. Moradi, B. Honarbakhsh, and T. F. Eibert, "Some Excitation Independent Bounds for the Total Active Reflection Coefficient of Antenna Arrays," *IEEE Transactions on Antennas and Propagation*, 2022.
- S. Moradi, B. Honarbakhsh, and T. F. Eibert, "Reply to the Comments on 'Some Excitation Independent Bounds for the Total Active Reflection Coefficient of Antenna Arrays'," *IEEE Transactions on Antennas and Propagation*, 2023.
- M. Manteghi and S. Moradi, "Front-End Mismatching, Mutual Coupling, Bandwidth, Transmission Line Noise, and SNR," *EuCAP*, 2024.
- S. Moradi, E. Mehrshahi, and J. A. Nossek, "Physical Elimination of Decoupling and Matching Networks in Multiport Antenna Arrays," *IEEE AP-S/URSI*, 2025.
- S. Moradi, E. Mehrshahi, T. F. Eibert, and J. A. Nossek, "Constrained Source Impedance and Excitation Optimization for Beamforming in Mutually Coupled Antenna Arrays Without Decoupling Networks," *under review at IEEE Transactions on Antennas and Propagation*, 2026.

Honors & Additional Training

- Ranked 1st in both B.Sc. and M.Sc. studies
- Direct M.Sc. admission as brilliant talent; also admitted to Sharif University of Technology under brilliant-talent route before choosing the University of Tehran track
- National Elite Foundation member
- Top 0.1% in the Iranian nationwide undergraduate entrance exam
- **Silicon Photonics Design, Fabrication and Data Analysis**, Sharif University of Technology
- Intermediate C++ Programming certificate

Languages

Persian (native), English (C2 / TOEFL iBT), German (A2)